



Question Set

Question 1

Consider a contingent claim priced by replication, risk-neutral probability, and change of numeraire. Define the state variables (state prices, density process, numeraire, and equivalent martingale measure) and derive the Radon-Nikodym pricing identity. State the measure, filtration, and tradability assumptions.

$$E^Q[X | \mathcal{F}_t] = \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t]$$

Question 2

For a contingent claim priced by replication, risk-neutral probability, and change of numeraire, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a contingent claim priced by replication, risk-neutral probability, and change of numeraire. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using a two-period tree with physical and risk-neutral probabilities, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For a contingent claim priced by replication, risk-neutral probability, and change of numeraire, compute and interpret replicating portfolio weights across states. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes state prices implied by tradable Arrow-Debreu claims. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a contingent claim priced by replication, risk-neutral probability, and change of numeraire. Explain how no-arbitrage bounds and terminal payoff consistency prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for a contingent claim priced by replication, risk-neutral probability, and change of numeraire. Specify measure-change weights applied to terminal claims, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a contingent claim priced by replication, risk-neutral probability, and change of numeraire and gets a stable but wrong answer. Explain how normalizing density weights incorrectly creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a contingent claim priced by replication, risk-neutral probability, and change of numeraire. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of a contingent claim priced by replication, risk-neutral probability, and change of numeraire, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of a two-period tree with physical and risk-neutral probabilities. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a contingent claim priced by replication, risk-neutral probability, and change of numeraire. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a contingent claim priced by replication, risk-neutral probability, and change of numeraire. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a contingent claim priced by replication, risk-neutral probability, and change of numeraire is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for state prices implied by tradable Arrow-Debreu claims are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

$$E^Q[X | \mathcal{F}_t] = \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t]$$

Question 17

For a contingent claim priced by replication, risk-neutral probability, and change of numeraire, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for measure-change weights applied to terminal claims. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that normalizing density weights incorrectly could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

$$\mathbb{E}^Q[X \mid \mathcal{F}_t] = \frac{1}{Z_t} \mathbb{E}^P[Z_T X \mid \mathcal{F}_t]$$

Question 20

Explain a contingent claim priced by replication, risk-neutral probability, and change of numeraire and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.